

Statement on Ethics Approval and Data Governance

Supporting Document for IEEE TMI Submission

Calibration-Free Non-Rigid CT-to-Bronchoscopy Registration for Image-Guided Biopsy
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1. Study type

This is a retrospective secondary analysis of existing, de-identified clinical imaging data (preoperative CT volumes and intra-operative bronchoscopy video frames). The study involves no prospective enrolment of participants, no research-directed intervention, no animal experiments, and no collection of identifiable private information.

2. Source and custodianship of the clinical data

The clinical CT-to-bronchoscopy data were acquired at Henan Provincial People’s Hospital (Zhengzhou, China) in the course of routine clinical care, and were de-identified by the responsible hospital department through its standard institutional review procedure prior to any research access. The data were made available to the research team by Dr. Nan Wei (co-author, Henan Provincial People’s Hospital).

3. Ethics approval / oversight

- **Institutional review.** Data access and use for this retrospective analysis were reviewed and authorised through the institutional data-governance and ethics-review process of Henan Provincial People’s Hospital.
- **Approval / waiver reference:** Not applicable — retrospective analysis of de-identified data conducted under the institutional data-governance authorisation of Henan Provincial People’s Hospital.
- **Informed consent.** Requirement for individual informed consent was not required, as the study is a retrospective analysis of de-identified data with no participant contact or intervention.
- **Declaration of Helsinki.** The study was conducted in accordance with the ethical principles of the Declaration of Helsinki (2013 revision).

4. Public benchmark data

Publicly released benchmarks used for validation (BOP Linemod/T-LESS, SCARED, the real-colon FEM benchmark, EndoNeRF, BlendedMVS) were used under their original providers’ access and

licence terms. These resources contain no new human or animal material acquired for the present study.

5. Data availability

The de-identified clinical imaging data are subject to the data-governance policy of Henan Provincial People's Hospital and are not publicly redistributable. Derived evaluation scripts and the software pipeline are released by the authors for reproducibility, as described in the manuscript.

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